

Siddhi Sanjay More

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Summary

MS Robotics candidate at Northeastern (Global Student Award) with hands-on experience across embedded systems, mechanical design and hardware. Comfortable across full stack: KiCAD schematics, Firmware on ESP32, CAD in SolidWorks and Dynamic simulation and analysis.

Technical Skills

Mechanical Systems and Fabrication: Mechanical Assemblies, Structural Analysis, Materials Selection, Tolerancing, Prototyping, Failure analysis, Laser and Foam cutting, FDM Printing, SLA Resin, Soldering (SMD and Through-hole), PCB Milling, Hand and Power Tools, Wiring

CAD and Analysis: SolidWorks, Fusion 360, CATIA, AutoCAD, NX 11, Ansys Workbench, Cero, DOE and ANOVA

Embedded Systems: KiCAD, ESP32 series, Arduino, IMUs (MPU 6050), FSRs, CAN-FD, I2C, L298N, Microcontrollers, real-time systems, Firmware Dev, Hardware Debugging, EasyEDA

Robotics: ROS/ROS2, Nav2, SLAM, Impedance Control, Gait Phase Estimation, Kinematics and IK, Trajectory Control, GCC-PHAT TDOA, FFT, Robo-analyzer, Moteus C1

Testing and Validation: Sensor Calibration, Odometry Analysis, Prototype Validation, System Debugging, Circuit Analysis

Programming: Python, C, MATLAB/Simulink, Linux, Git, Arduino IDE, TensorFlow/Keras

Education

MS in Robotics, *Northeastern University, Boston, MA*

Expected May 2027

Concentration: Mechanical Engineering, **GPA:** 3.25, **Academic Awards:** Global Student Award

Courses: Robot Sensing & Navigation, Robotics Mechanics & Control, Assistive Robotics, Wearable Robotics, Control Systems Engineering, Foundations of AI

BE in Mechanical Engineering, *Mumbai University, India*

Jul 2020 – Jul 2024

Honors: Robotics, **Courses:** FEA, Machine Design, Automation & AI, CAD/CAM, IoT, Supply Chain Management

Projects

Smart Ankle Prosthetics, *Northeastern University*

Jan 2026 – Present

- Wired an ESP32-S2 Mini with MPU6050 and FSR sensors on the intact limb; synced ankle angle and heel/toe pressure streams in real time and logged to CSV via Python for offline analysis
- Built a MATLAB gait phase estimator that classifies heel-strike, mid-stance, toe-off, and swing from fused IMU/FSR data that predicts the next phase 75 ms ahead so the prosthetic ankle pre-positions before weight transfer
- Architected CAN-FD messaging to a Moteus C1 motor controller, sending phase suggestions rather than raw commands that keeps adaptive torque response in the ankle controller rather than forcing it from the host. Currently resolving FSR comms

Semi-Autonomous Aquatic Cleanup Robot, *Smart India Hackathon*

Jul 2023 – Oct 2023

- Designed the chassis, hull geometry, and mechanical crusher system in SolidWorks; sized scooping geometry for varying hyacinth densities, modeled intake/drainage systems for water flow management, and validated crusher prototype against target throughput
- Co-wrote ESP32-CAM firmware for real-time object detection and obstacle avoidance; integrated an ultrasonic sensor suite and Arduino Mega to close the perception-to-actuation loop in aquatic conditions
- Trained a CNN in TensorFlow/Keras on ~1,000 labelled images of water hyacinth hitting 87.5% classification accuracy, feeding detections into the autonomous navigation logic

Plastic Injection Molding Optimization via FEA & DOE, Mumbai University**Jul 2023 – Oct 2024**

- Modelled automotive airbag housings in SolidWorks Plastics and ran a full Design of Experiments to isolate process parameters driving defects, reducing sink marks by 20% and warpage by 11%
- Built an ANOVA framework on top of the DOE results to rank parameter significance, giving the team a clear priority order for process changes without running exhaustive physical trials
- Published findings as “Experimentation in Injection Molding: Simulation and Parameter Optimization Using DOE and ANOVA” in AIP Conference Proceedings, ICAPSM 2024

Experience

Robotics Intern, IIT Bombay & IStop.ai, India**Feb 2022 – Apr 2022**

- Derived DH parameters and built forward/IK solvers for a SCARA robot in MATLAB, with end-effector accuracy landing within sub-mm across the full reachable workspace
- Performed FMEA under max-payload static load cases in ANSYS Workbench; identified a critical stress concentration and triggered a design revision before fabrication.
- Validated joint-to-end-effector motion profiles in Robo-Analyzer, cutting debug cycles and shaving time off controller tuning

CAD Design Intern, IIT Bombay & IStop.ai, India**Feb 2022 – Apr 2022**

- Designed industrial mechanical parts in SolidWorks against real client specifications; submitted completed models as a formal deliverable and received a completion certificate
- Applied GD&T and manufacturing tolerancing standards throughout part design, working within constraints set by fabrication process requirements